

Quiz 5: Solutions

Give exact answers and justify your reasoning. Do not use electronic tools. All functions are real-valued, and angles are measured in radians. Limits are taken within the stated domain. You may use results covered through Lecture 6, including the derivative definition and the calculation of the derivative of x^2 . Do not use differentiation rules or L'Hospital's rule.

1. Prove directly from the definition that

$$\lim_{x \rightarrow 1} \frac{2 + \sin x}{(x - 1)^2} = +\infty.$$

Give an explicit δ for every $M > 0$.

Solution.

Since $-1 \leq \sin x \leq 1$, we have $2 + \sin x \geq 1$. Given $M > 0$, choose

$$\delta = \frac{1}{\sqrt{M}}.$$

If $0 < |x - 1| < \delta$, then

$$\frac{2 + \sin x}{(x - 1)^2} \geq \frac{1}{(x - 1)^2} > \frac{1}{\delta^2} = M.$$

This proves the limit $+\infty$ directly from its definition.

2. Let $f(x) = \sqrt{x}$, with domain $[0, \infty)$. Use the definition of the derivative to calculate $f'(4)$. Then give the equation of the tangent line at $(4, 2)$.

Solution.

For $0 < |h| < 1$, we have $4 + h > 0$, and rationalization gives

$$\frac{f(4 + h) - f(4)}{h} = \frac{\sqrt{4 + h} - 2}{h} = \frac{1}{\sqrt{4 + h} + 2}.$$

Root continuity and the quotient law yield

$$f'(4) = \lim_{h \rightarrow 0} \frac{f(4 + h) - f(4)}{h} = \frac{1}{4}.$$

The tangent line through $(4, 2)$ therefore has equation

$$y = 2 + \frac{1}{4}(x - 4) = \frac{x}{4} + 1.$$